

Effectiveness of palliative care for the management of end-stage liver disease: a systematic review protocol

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ABSTRACT

Objective: The aim of this systematic review will be to evaluate the effectiveness of palliative care for treating adults with end-stage liver disease (ESLD), focusing on outcomes such as quality of life, quality of care, symptom burden, and patient satisfaction.

Introduction: ESLD is a critical health issue, characterized by various debilitating symptoms that negatively affect quality of life for patients. Palliative care, which is a specialty used to alleviate pain and control other manifestations of the disease, has been suggested across various studies for the management of quality of life and symptom burden in patients with ESLD.

Eligibility criteria: This review will include studies focusing on the use of palliative care for adults (≥ 18 years) with ESLD. Papers that compare palliative care with usual care or placebo will also be included.

Methods: The review will follow the JBI methodology for systematic reviews of effectiveness. PubMed, Embase (Ovid), Cochrane Central Register of Controlled Trials (CENTRAL), CINAHL (EBSCOhost), Scopus, and Web of Science will be searched for published studies. Sources to be searched for unpublished literature will include Google Scholar, ProQuest Dissertations and Theses Global, Shodhganga, MedNar, OpenGrey, and the WHO International Clinical Trials Registry Platform. Two independent reviewers will assess titles, abstracts, and full-text papers against the inclusion criteria. A third reviewer will resolve any discrepancies. Standardized JBI appraisal tools will be used to critically appraise eligible studies. Two independent reviewers will extract data. Meta-analysis will be conducted for outcomes, where feasible. If a meta-analysis cannot be conducted, the findings will be reported in a narrative format, adhering to the Synthesis Without Meta-analysis guidelines. Certainty of evidence will be reported using the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach.

Review registration: PROSPERO CRD42024586855

Keywords: end-stage liver disease (ESLD); palliative care; quality of care; quality of life

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Introduction

End-stage liver disease (ESLD) is becoming a leading cause of death, with around 2 million people dying from the disease annually worldwide.^{1,2} ESLD has been highlighted as a significant health issue by the

World Health Organization due to the disease's progression, its complications, and symptom management.

The causes of ESLD include alcoholic liver disease, viral hepatitis, non-alcoholic fatty liver disease (NAFLD), autoimmune liver diseases, and hepatocellular carcinoma, among others.³ These diseases cause an increase in disease burden globally, which varies by region. In Africa and Asia, viral hepatitis is the leading cause of increased ESLD, whereas NAFLD and alcoholic liver disease are rising in high-income countries such as the USA and Europe.^{2,4} This raises the issue of

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public health importance, as increased global prevalence indicates higher chances of mortality and symptom burden. The resultant impaired quality of life needs integrated care approaches.⁵ This also has a negative economic impact on these patients with ESLD, as increased hospitalization results in loss of jobs and financial burden.⁶

Illnesses such as ESLD decrease health-related quality of life.⁷ The complications of ESLD include cirrhosis with portal hypertension, ascites, spontaneous bacterial peritonitis, hepatic encephalopathy, viral hepatitis jaundice, and variceal bleeding.⁸ These complications need different management modalities such as nutritional therapy,⁹ psychosocial support, and an interdisciplinary approach involving palliative care¹⁰ that addresses physical, psychosocial, and emotional requirements. It is projected that by 2060, about 48 million people will be affected and die from serious health-related suffering each year. With this increase, palliative care in serious illnesses and end-of-life care is necessary. This is essential for symptom management and dealing with psychological and social aspects of the disease as well as the suffering associated with it.⁷ The only other definitive cure for ESLD is liver transplantation, which is only feasible for a few. Of those candidates who are eligible for transplantation, about 20% do not make it through the waiting period.¹ Early implementation of care and treatment plans have been demonstrated to be advantageous but are rarely administered by physicians for patients with ESLD.¹¹

In such vulnerable patients, palliative care is one of the treatments that is recommended.¹⁰ It has been found that palliative care improves the quality of life and alleviates symptoms of the disease while having positive effects on caregivers.¹⁰ *Palliative care*, as defined by the International Association for Hospice and Palliative Care, is the active, comprehensive care of people of all ages experiencing significant health-related distress, especially individuals nearing the end of life,¹² while focusing on meeting the physical, spiritual, and psychosocial needs of these patients.¹¹ Palliative care can be delivered through various consolidated approaches such as pharmacological, non-pharmacological, and cognitive behavior therapy,¹³ mindfulness exercises, acceptance and commitment therapy,¹⁴ and social support interventions such as end-of-life discussions and family counseling.¹⁰

Palliative care is best implemented by employing an interdisciplinary approach that involves hepatologists,

nurses, social workers, and palliative care specialists whose cooperation guarantees holistic care for patients with ESLD.¹⁰ Nevertheless, the integration of early palliative care should be considered, as the consequential morbidity and mortality related to ESLD necessitates its timely utilization.¹⁵ Introducing palliative care early in the course of advanced chronic disease has demonstrated an improvement in the quality of life¹⁶ of patients and their caregivers.¹ Palliative care helps improve physical, psychological, and social symptom burden. It can help with spiritual distress in patients,¹ along with psychological and financial issues¹ while also setting patient-based goals.^{10,17,18} Despite the projection of a poor prognosis and substantial symptom burden of ESLD, most research indicates that palliative care is infrequently incorporated and typically utilized only after patients are excluded from the liver transplantation list or during their end-of-life.¹¹

Symptom management, enhanced communication, and decision-making with coordinated care and patient-centered approaches provided by palliative care help improve the quality of care. Our preliminary search on PubMed and the Cochrane Database of Systematic Reviews identified 2 systematic reviews conducted on palliative care as an intervention in patients with ESLD. Vijeratnam *et al.* conducted a systematic review of how palliative care is applied in patients with ESLD who are waiting for liver transplantation.¹⁹ This review mainly focused on adapting specialist palliative care in such patients and the perspectives of health care professionals. A review by Das *et al.* consisted of experiences, perspectives, and expectations of patients diagnosed with advanced liver disease on the usage of palliative care.²⁰ However, none of the reviews analyzed the effectiveness of palliative care in patients with ESLD for improving primary and secondary outcomes, such as quality of life, quality of care, symptom burden, and patient satisfaction.

Although palliative care is increasingly acknowledged as crucial for ESLD management, substantial knowledge gaps persist regarding its efficiency and integration. There is a gap in understanding how hospital-based and home-based palliative care affects patients' quality of life, symptom burden, and satisfaction. The lack of evidence has significant implications, potentially leading to underutilization of palliative care interventions and suboptimal symptom management in patients with ESLD.¹⁵ The delayed introduction of palliative care limits its effectiveness for addressing

patient needs, such as increased symptom burden leading to increased hospitalizations.^{13,16} Moreover, it is necessary to ensure that the care provided is aligned with the patient's treatment preferences and goals,¹³ as there is often uncertainty about the disease trajectory.¹⁰ This review is expected to enhance the understanding of palliative care approaches, informing clinicians on optimal care strategies for this vulnerable population. The findings have significant implications for clinical practice, which will promote earlier access to palliative care and overall improvement in patient care while supporting optimized health outcomes.

Review question

What is the effectiveness of palliative care for managing ESLD in hospital and home-based settings?

Eligibility criteria

Participants

This review will include studies involving adults aged 18 years or over diagnosed with ESLD, with evidence of at least one of the following: patients with liver cirrhosis referred for liver transplantation assessment; liver cirrhosis-related liver transplant candidates; patients with chronic or advanced liver disease being referred for liver transplantation; and patients with liver cirrhosis or advanced liver disease not eligible for liver transplantation. Patients diagnosed with non-hepatic cancer and those who have undergone previous liver transplantation will be excluded.

Intervention

This review will encompass studies examining the use of palliative care in the treatment of ESLD. Individual or a combination of hospital-based palliative care and home-based palliative care delivered by palliative care specialists, trained hepatologists and nurses, and trained home health aides, will be included.

Comparator

This review will consider studies that assess the intervention against hospital-based or home-based usual/standard care for ESLD or a placebo.

Outcomes

This review will focus on studies with the following primary outcomes:

- quality of life (defined as an individual's overall well-being, comfort, and ability to engage in

daily activities) using validated scales from primary papers, which include generic scales such as EQ-5D-5L and PROMIS-29, and disease-specific measurements such as the Chronic Liver Disease Questionnaire

- quality of care (defined as the extent to which health care services enhance the likelihood of achieving optimal outcomes with regard to an individual's or population's health) using scales such as the Palliative care Outcome Scale and the integrated Palliative care Outcome Scale (which uses a 5-point Likert scale for measurement).

Secondary outcomes will include:

- assessing symptom burden (defined as the impact of physical and psychological symptoms of a disease experienced by individuals) using validated scales such as the Edmonton Symptom Assessment System
- overall patient satisfaction (measured by descriptive questions or patient survey).

The outcomes in this review will be measured from baseline to 3 months of assessment. Further assessments of individual outcomes will be carried out at 6, 9, and 12 months to evaluate their sustainability effects. Studies with a minimum follow-up period of at least 2 months will be included in this review.

Types of studies

This review will primarily focus on experimental and quasi-experimental study designs, including randomized controlled trials (RCTs), non-RCTs, before-and-after studies, and interrupted time-series studies. Due to potential ethical constraints that may limit the feasibility of RCTs and non-RCTs, we will also consider analytical observational studies, including retrospective and prospective cohort studies, case-control studies, and analytical cross-sectional studies. The inclusion of these observational designs will allow for a more comprehensive understanding of the topic.

Methods

This review will be conducted following the JBI methodology for systematic reviews of effectiveness²¹ and will be reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist.²² This protocol has been registered in PROSPERO (CRD42024586855).

Search strategy

A comprehensive search will be performed to locate published and unpublished studies. A 3-step search strategy was undertaken: firstly, a preliminary PubMed search was conducted to identify pertinent articles followed by an analysis of title/abstract text words and index terms. The analysis was used to devise a comprehensive search strategy using Medical Subject Headings (MeSH) to find various entry terms for the keywords (see Appendix I). These search terms were concatenated incorporating the Boolean operators “OR” and “AND” to construct the search strings. Secondly, the search terms were adapted for utilization in multiple databases. Thirdly, the reference lists of all incorporated studies will undergo supplementary screening to identify additional relevant studies.

The databases to be searched will include PubMed, Embase (Ovid), Scopus, CINAHL (EBSCOhost), Web of Science, and Cochrane Central Register of Controlled Trials (CENTRAL). Sources of unpublished studies and gray literature will include Google Scholar, ProQuest Dissertations and Theses Global, Shodh-ganga, MedNar, OpenGrey, and the WHO International Clinical Trials Registry Platform. The search strategy for PubMed is provided in Appendix I.

To ensure a comprehensive understanding of the effectiveness of palliative care for managing ESLD, the literature search will be conducted without any specific time constraints. This approach will enable the inclusion of all relevant studies, provided they adhere to the inclusion criteria, to ensure that the review incorporates evolving methodologies and outcomes in palliative care. Studies published in languages other than English will be translated using DeepL software (DeepL, Cologne, Germany); however, due to time and financial constraints, the translations will not be validated.

Study selection

The search results will be integrated and imported into Zotero (Corporation for Digital Scholarship and Roy Rosenzweig Center for History and New Media, VA, USA), following which, duplicates will be identified and deleted. To manage the extensive number of results typically returned in databases such as Google Scholar and MedNar, only the first 20 pages of results will be imported for screening, as evidence suggests that most relevant studies are concentrated within this range.²³ A pilot will be undertaken for the first 10 studies obtained using our search strategy to assess the

clarity and feasibility of our inclusion and exclusion criteria. Titles and abstracts will be assessed against the inclusion criteria by at least 2 independent reviewers (AM, HM). Relevant studies will be retrieved in full and the citation details will be imported into the JBI System for the Unified Management, Assessment and Review of Information (JBI SUMARI; JBI, Adelaide, Australia).²⁴ Selected studies will undergo a full-text evaluation against the inclusion criteria by 2 reviewers (AM, HM).

For mixed-population studies, studies with patients diagnosed with ESLD as well as comorbidities but receiving palliative care will be included; however, these studies will not undergo meta-analysis with the other studies that do not include patients with comorbidities. Studies that focus on ESLD along with other diseases receiving palliative care will be excluded. The reasons for excluding studies at full-text will be reported in the review. Any disagreements that arise between the 2 reviewers will be resolved through a consensus-driven discussion or in consultation with an additional reviewer (DJ).

The final systematic review will include the search and study selection outcomes, which will be illustrated in a PRISMA flow diagram.²²

Assessment of methodological quality

Two independent reviewers (AM, HM) will conduct a methodological quality assessment of eligible studies using standardized JBI critical appraisal tools, modified for experimental,²⁵ quasi-experimental,²⁶ and observational studies.^{27,28} Authors will be contacted to obtain missing or supplementary data, as necessary, up to 3 times. Disagreements will be resolved through discussion or via a third reviewer (DJ). The critical appraisal findings will be presented in tabular format, supplemented with a narrative summary. Risk of bias from experimental, quasi-experimental, or observational studies will be assessed separately to guide the interpretation and synthesis of results from individual studies of each study design. Regardless of their methodological quality, all studies will undergo data extraction and synthesis, where feasible.

Data extraction

Data will be extracted independently by 2 reviewers (AM, HM) from papers included in the review using the standard data extraction tool available in JBI SUMARI,²⁴ which will be modified for this review (see Appendix II). Extracted data will include study

design, participant demographics, interventions, and outcome measures aligned with the review objectives. Any discrepancies will be resolved through discussion or with a third reviewer (DJ). Authors will be approached to request missing or supplementary data, when required. If no response is received, a follow-up email will be sent after 2 weeks, and if no response is received after a further 2 weeks, the study will be excluded.

Data synthesis

Where feasible, a statistical meta-analysis will be conducted using JBI SUMARI.²⁴ The study will present effect sizes using relative risk, odds ratio for dichotomous outcomes, and weighted or standardized mean differences, and standard deviation for continuous data, all with 95% CIs. Weighted mean differences will be used for continuous data measured using a uniform scale; standardized mean differences will be calculated for heterogeneous scale data.²⁹ Authors shall be contacted for further information or guidance on calculations, which will adhere to the Cochrane Handbook³⁰ if the standard deviation is not reported. When both methods are not possible, the studies with the higher standard deviation will be incorporated into the meta-analysis. In cases of duplicated publications, authors will be contacted to clarify data discrepancies. In such situations, priority will be given to studies with larger sample sizes or superior methodological quality.

The random effects model will be used to perform statistical analyses. If fewer than 5 studies are included in the meta-analysis, the fixed effects model will be used, where appropriate. Subgroup analyses will be conducted where sufficient studies use the same methodology and study design, or sufficient data to investigate the effect of type of palliative care, gender, and age on quality of life or care in patients with ESLD. A sensitivity analysis will be conducted to determine the impact of the methodological quality (ie, to include or exclude poor-quality studies from the meta-analysis) and sample size (ie, to include or exclude studies with large or small sample sizes).

Heterogeneity will be evaluated using forest plots, Cochran's Q statistic, and the I^2 statistic. For the Q statistic, a significance level of <0.10 will be employed (instead of <0.056) when fewer than 20 studies are available. Variations among studies caused by true heterogeneity will be categorized as low, moderate, or high (I^2 values of 25%, 50%, or 75%, respectively).

The appropriateness of reporting meta-analysis results will be discussed among reviewers if the I^2 value exceeds 50%. Caution will be exercised in interpreting both Q and I^2 statistics when the number of studies is fewer than 10.

To assess publication bias, a funnel plot will be generated if there are ≥ 10 studies included in the meta-analysis. Statistical tests for funnel plot asymmetry (Egger test) will be performed where appropriate.

If a meta-analysis cannot be performed, a narrative summary of results will be provided, supplemented with illustrative tables and figures. The narrative findings will adhere to the Synthesis Without Meta-analysis (SWiM) guidelines.³¹

Assessing certainty in the findings

The Grading of Recommendations, Assessment, Development and Evaluation (GRADE)³² approach will be employed to assess the certainty of evidence. A Summary of Findings will be generated using GRADEpro GDT (McMaster University, ON, Canada). This will be undertaken for each outcome by 2 independent reviewers (AM, HM). Any reviewer disagreements will be resolved through discussion or with a third reviewer (DJ). The authors of papers will be contacted to request missing or additional data for clarification, where required.

The Summary of Findings will present the following information where applicable: absolute risks for both the intervention and comparator, estimates of relative risk, and a ranking of the quality of evidence. The ranking of the quality of evidence will be based on several factors, including the risk of bias, directness of evidence, heterogeneity, precision, and publication bias that poses a risk to the validity of the review findings. The outcomes reported in the Summary of Findings will be: quality of life, quality of care, symptom burden, and patient satisfaction.

To ensure the consistency and reproducibility of the GRADE assessments between reviewers, the 2 reviewers (AM and HM) conducting the assessment will be trained by the third reviewer (DJ). Additionally, the checklist developed by Meader *et al.*³³ will guide the reviewers in extracting the data required to make the GRADE assessments.

Author contributions

AM and HM jointly developed the review question and formulated the protocol. BS worked on the search

strategy and inclusion criteria. SR supervised the protocol. CV offered subject-specific expertise on palliative care. DJ provided methodological expertise and conceptualized the review topic. All authors reviewed and edited the draft and approved the final submission.

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Appendix I: Search strategy

PubMed		
Search conducted: August 8, 2024		
Search terms	MeSH	Text words
Palliative care	Palliative Care [MeSH]	Palliative Care [tiab]
End stage liver disease	End stage liver disease [MeSH]	End stage liver disease [tiab], ESLD [tiab]
Liver cirrhosis	Liver cirrhosis [MeSH]	Liver cirrhosis [tiab]
Portal hypertension	Portal hypertension [MeSH]	Portal hypertension [tiab]
Variceal bleeding	Variceal bleeding [MeSH]	Variceal bleeding [tiab]
Hepatocellular carcinoma	Hepatocellular carcinoma [MeSH]	Hepatocellular carcinoma [tiab]
Liver failure	Liver failure [MeSH]	Liver failure [tiab]
Jaundice	Jaundice [MeSH]	Jaundice [tiab]
Quality of life	Quality of Life [MeSH]	Quality of Life [tiab], QoL [tiab], HRQOL [tiab]
Quality of care	Quality of Health Care [MeSH]	Quality of Care [tiab], QoC [tiab], Quality of Health Care [tiab]
Patient satisfaction	Patient Satisfaction [MeSH]	Patient Satisfaction [tiab]
Symptom burden	Symptom burden [MeSH]	Symptom burden [tiab]
Fatigue	Fatigue [MeSH]	Fatigue [tiab]
Muscle fatigue	Muscle fatigue [MeSH]	Muscle fatigue [tiab]
Chronic fatigue syndrome	Fatigue syndrome, chronic [MeSH]	Fatigue syndrome, chronic [MeSH]
Pain	Pain [MeSH]	Pain [tiab]
Sleepiness	Sleepiness [MeSH]	Sleepiness [tiab]
Anxiety	Anxiety [MeSH]	Anxiety [tiab]
Dyspnea	Dyspnea [MeSH]	Dyspnea [tiab]
Depression	Depression [MeSH]	Depression [tiab]
Nausea	Nausea [MeSH]	Nausea [tiab]
Muscle cramps	Muscle cramps [MeSH]	Muscle cramps [tiab]

PubMed		
Search conducted: August 8, 2024		
Sl no.	Query	Records retrieved
#1	(Palliative Care [MeSH]) OR (Palliative Care [tiab])	84,721
#2	(End stage liver disease [MeSH]) OR (End stage liver disease [tiab]) OR (ESLD [tiab])	14,849
#3	((#1) AND (#2))	155
#4	(Liver cirrhosis [MeSH]) OR (Liver cirrhosis [tiab]) OR (Portal hypertension [MeSH]) OR (Portal hypertension [tiab]) OR (Variceal bleeding [MeSH]) OR (Variceal bleeding [tiab]) OR (Hepatocellular carcinoma [MeSH]) OR (Hepatocellular carcinoma [tiab]) OR (liver failure [MeSH]) OR (Liver failure [tiab]) OR (jaundice [MeSH]) OR (Jaundice [tiab])	3,64,554
#5	((#3) AND (#4))	122
#6	(Quality of Life [MeSH]) OR (Quality of Life [tiab]) OR (QoL [tiab]) OR (HRQOL [tiab]) OR (Quality of Health Care [MeSH]) OR (Quality of Care [tiab]) OR (QoC [tiab]) OR (Quality of Health Care [tiab]) OR (Patient Satisfaction [MeSH]) OR (Patient Satisfaction [tiab]) OR (Symptom burden [MeSH]) OR (Symptom burden [tiab]) OR (Fatigue [MeSH]) OR (Fatigue [tiab]) OR (Muscle Fatigue [MeSH]) OR (Muscle Fatigue [tiab]) OR (Fatigue Syndrome, Chronic [MeSH]) OR (Fatigue Syndrome, Chronic [tiab]) OR (Pain [MeSH]) OR (Pain [tiab]) OR (Sleepiness [MeSH]) OR (Sleepiness [tiab]) OR (Anxiety [MeSH]) OR (Anxiety [tiab]) OR (Dyspnea [MeSH]) OR (Dyspnea [tiab]) OR (Depression [MeSH]) OR (Depression [tiab]) OR (Nausea [MeSH]) OR (Nausea [tiab]) OR (Muscle cramps [MeSH]) OR (Muscle cramps [tiab])	99,20,525
#7	#5 AND #6	92

Appendix II: Draft data extraction form

Study	
Lead author	
Year of publication	
Country	
Single or Multicenter	
Study design	
Randomization	
Blinding	
Confounders	
Inclusion criteria	
Exclusion criteria	
Population characteristics	
Sample size	
Intervention	n=
Control	n=
Losses to follow-up	
Stopped trial early	
Intervention 1	
Setting 1: home or hospital based	
Setting 2: integrated with standard care	
Intervention 2: placebo or standard care	
Quality of life (outcome 1) details	
Outcome 1 intervention 1	
Outcome 1 intervention 2	
Quality of care (outcome 2) details	
Outcome 2 intervention 1	
Outcome 2 intervention 2	
Symptom burden (outcome 3) details	
Outcome 3 intervention 1	
Outcome 3 intervention 2	
Patient satisfaction (outcome 4) details	
Outcome 4 intervention 1	
Outcome 4 intervention 2	
Reviewer comments	
Sponsorship/funding	